AI 3000: Reinforcement Learning
Tutorial 2

1 One-Period MDP Problem

Consider the following one-Period MDP problem:

- States: $S = \{s_0, s_1, s_2\}$ with the state s_0 being the initial state, and s_1, s_2 being absorbing states.
- Actions: $A(s_0) = \{a_1, a_2, a_3, a_4, a_5\}$

and the one-step transition probabilities and rewards being defined as follows:

Action	$P(s_1 \mid s_0, a)$	$P(s_2 \mid s_0, a)$	$R(s_0, a, s_1)$	$R(s_0, a, s_2)$
a_1	0.6	0.4	+10	+0
a_2	0.5	0.5	+8	+0
a_3	0.9	0.1	+2	+0
a_4	0.2	0.8	+14	+4
a_5	0.3	0.7	+5	+2

1. Find the state-action values for all the available actions.
2. Find the set of all optimal policies for the setting given above. How many such policies exist?

2 Finite Horizon MDPs

Consider a Finite-Horizon Markov Decision Process (MDP) defined by the following parameters:

- **State Space:** $S = \{s_1, s_2\}$
- **Action Spaces:** $A_{s_1} = \{a_{1,1}, a_{1,2}\}$ and $A_{s_2} = \{a_{2,1}, a_{2,2}\}$
- **Transition Probabilities:** For $n = 1, 2$:

$$\begin{aligned} p_n(s_1 \mid s_1, a_{1,1}) &= 0.8, & p_n(s_2 \mid s_1, a_{1,1}) &= 0.2 \\ p_n(s_1 \mid s_1, a_{1,2}) &= 0, & p_n(s_2 \mid s_1, a_{1,2}) &= 1.0 \\ p_n(s_1 \mid s_2, a_{2,1}) &= 0, & p_n(s_2 \mid s_2, a_{2,1}) &= 1.0 \\ p_n(s_1 \mid s_2, a_{2,2}) &= 0.4, & p_n(s_2 \mid s_2, a_{2,2}) &= 0.6 \end{aligned}$$

- **Rewards:** For $n = 1, 2$:

$$\begin{aligned} r_n(s_1, a_{1,1}, s_1) &= 5, & r_n(s_1, a_{1,1}, s_2) &= 5 \\ r_n(s_1, a_{1,2}, s_1) &= 0, & r_n(s_1, a_{1,2}, s_2) &= 5 \\ r_n(s_2, a_{2,1}, s_1) &= 0, & r_n(s_2, a_{2,1}, s_2) &= 5 \\ r_n(s_2, a_{2,2}, s_1) &= 20, & r_n(s_2, a_{2,2}, s_2) &= 10 \end{aligned}$$

Analyse the above MDP, with horizon $N = 2$ and calculate the expected total reward generated by the following policies when starting from initial state s_1.

2.1 (Markovian Deterministic Policy) Let $\pi_1 = (d_1, d_2)$ be a Markovian deterministic policy where the decision rules d_1 and d_2 are defined as follows:

$$d_1(s) = \begin{cases} a_{1,1}, & \text{if } s = s_1 \\ a_{2,1}, & \text{if } s = s_2 \end{cases} \quad \text{and} \quad d_2(s) = \begin{cases} a_{1,2}, & \text{if } s = s_1 \\ a_{2,1}, & \text{if } s = s_2 \end{cases}$$

2.2 (Markovian Randomized Policy) Let $\pi_2 = (d_1, d_2)$ be a Markovian randomized policy where the decision rules are defined as follows:

$$w_{d_n}(a \mid s) = \begin{cases} q_1^n, & a = a_{1,1},\, s = s_1 \\ 1 - q_1^n, & a = a_{1,2},\, s = s_1 \\ q_2^n, & a = a_{2,1},\, s = s_2 \\ 1 - q_2^n, & a = a_{2,2},\, s = s_2 \end{cases}$$

where, $q_1^1 = 0.1, q_2^1 = 0.5\, q_1^2 = 0.5, q_2^2 = 0.7$.

2.3 (History-Dependent Deterministic Policy) Consider the policy $\pi_3 = (d_1, d_2)$, where the decision rules at each time-step are defined as follows:

$$d_1(h_1) = \begin{cases} a_{1,1}, & \text{if } h_1 = s_1 \\ a_{2,1}, & \text{if } h_1 = s_2 \end{cases} \quad \text{and} \quad d_2(h_2) = \begin{cases} a_{1,2}, & \text{if } h_2 = (s_1, a_{1,1}, s_1) \\ a_{2,1}, & \text{if } h_2 = (s_1, a_{1,1}, s_2) \\ a_{2,2}, & \text{if } h_2 = (s_2, a_{2,1}, s_2) \end{cases}$$

where h_i gives us the history up to time-step i

2.4 (History-Dependent Randomized Policy) The policy $\pi_4 = (d_1, d_2)$, where the decision rules at each time-step are defined as follows:

- At the first decision step, the randomized decision rule d_1 is given by:

$$w_{d_1}(a \mid h) = \begin{cases} q_1^1, & a = a_{1,1},\, h = s_1 \\ 1 - q_1^1, & a = a_{1,2},\, h = s_1 \\ q_2^1, & a = a_{2,1},\, h = s_2 \\ 1 - q_2^1, & a = a_{2,2},\, h = s_2 \end{cases}$$

  where $q_1^1 = 0.1, q_2^1 = 0.5$

- For the second decision step, let the following be the possible histories:

$$\begin{aligned} h_1 &= (s_1, a_{1,1}, s_1), & h_4 &= (s_2, a_{2,1}, s_2) \\ h_2 &= (s_1, a_{1,1}, s_2), & h_5 &= (s_2, a_{2,2}, s_1) \\ h_3 &= (s_1, a_{1,2}, s_2), & h_6 &= (s_2, a_{2,2}, s_2) \end{aligned}$$

  And the randomized decision rule d_2 depends on the history h_i as follows:

  – For histories ending in state s_1:

$$w_{d_2}(a \mid h_i) = \begin{cases} q_i^2, & a = a_{1,1} \\ 1 - q_i^2, & a = a_{1,2} \end{cases}$$

– For histories ending in state s_2:

$$w_{d_2}^2(a \mid h_i) = \begin{cases} q_i^2, & a = a_{2,1} \\ 1 - q_i^2, & a = a_{2,2} \end{cases}$$

here, $q_1^2 = 0.4, q_2^2 = 0.6, q_3^2 = 0.4, q_4^2 = 0.6, q_5^2 = 0.5, q_6^2 = 0.5$

3 Policy-Value Function

Consider a simplified version of the Recycling Robot discussed in lecture 2, with the MDP defined as follows

- States: High(H) and Low(L)
- Actions: Search(s) and Recharge(r)

with the one-step transition probabilities and rewards being defined as follows:

- In State H:
  - Action Search: 80% chance to stay in H (Reward +10), 20% chance to drop to L (Reward +10).
  - Action Recharge: Waste of energy; stays in H (Reward 0).
- In State L:
  - Action Search: Risk of draining battery completely! 30% chance to stay in L (Reward +3), 70% chance battery dies (reset to H with penalty Reward −5).
  - Action Recharge: Deterministically moves back to H (Reward 0).

Now consider the following stationary policies:

1. The robot always searches ,i.e., $\pi(s \mid H) = \pi(s \mid L) = 1$
2. The robot is less risk-taking when in a low charge state, i.e., $\pi(s \mid H) = 0.9$ and $\pi(s \mid L) = 0.4$

Compare the value functions of these policies for an expected discounted reward criterion with a 2 step horizon with the discount factor being $\gamma = 0.1$ and 0.9